1.37 Integration: Definite Integrals and Areas

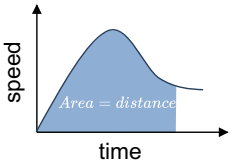
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Definite Integration

19

So far we've seen integration as '**the opposite of differentiation**', allowing us to find $y = f(x)$ when we know the gradient function $y = f'(x)$.

In practical settings however the most useful use of integration is that **it finds the area under a graph**. Remember at GCSE for example when you estimated the area under a speed-time graph, using trapeziums, to get the distance?
If you knew the equation of the curve, you could get the exact area!

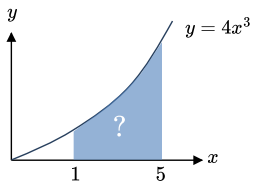


Before we do this, we need to understand how to find a **definite integral**:

These are known as **limits**, which give the values of x we're finding the area between.

We integrate as normal, but put expression in **square brackets**, meaning **we still need to evaluate the integrated expression using the limits**.

$$\int_1^5 4x^3\,dx =$$



Write $(\ldots) - (\ldots)$ and evaluate the expression for each of the limits, top one first.

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Another Example

20

You can use the $\left[\!\left[\int_b^a \square\right]\!\right]$ button on your calculator to evaluate definite integrals.
But only use it to check your answer.

1.37 Integration: Definite Integrals and Areas

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Problem Solving

21

Given that P is a constant and $\int_1^5 (2Px + 7)dx = 4P^2$, show that there are two possible values for P and find these values.

Remember: P is a constant, so just treat it as a number.

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Exercise 13D – Y1 page 297

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4 Given that A is a constant and $\int_1^4 (6\sqrt{x} - A)\,dx = A^2$, show that there are two possible values for A and find these values.

1.37 Integration: Definite Integrals and Areas

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Exercise 13D – Y1 page 297

23

5 Use calculus to find the value of $\int_1^9 (2x - 3\sqrt{x})\,\mathrm{d}x$.

6 Evaluate $\int_4^{12} \frac{2}{\sqrt{x}}\,\mathrm{d}x$, giving your answer in the form $a + b\sqrt{3}$, where a and b are integers.

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Exercise 13D – Y1 page 297

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7 Given that $\int_1^k \frac{1}{\sqrt{x}}\,\mathrm{d}x = 3$, calculate the value of k.

1.37 Integration: Definite Integrals and Areas

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EXTENSION

25

1 [MAT 2009 1A] The smallest value of

$$I(a) = \int_0^1 (x^2 - a)^2 \, dx$$

as a varies, is what?

2 [MAT 2015 1D] Let

$f(x) = \int_0^1 (xt)^2 \, dt$ and $g(x) = \int_0^x t^2 \, dt$

Let $A > 0$. Which of the following statements are true?

$g(f(A))$ is always bigger than $f(g(A))$

$f(g(A))$ is always bigger than $g(f(A))$

They are always equal.

$f(g(A))$ is bigger if $A < 1$, and $g(f(A))$ is bigger if $A > 1$.

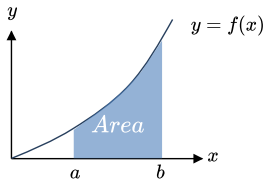
$g(f(A))$ is bigger if $A < 1$, and $f(g(A))$ is bigger if $A > 1$.

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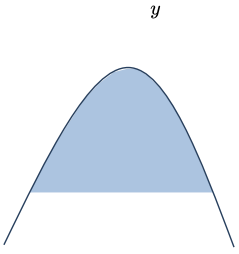
Areas under curves

26

Earlier we saw that the definite integral $\int_a^b \mathbf{f(x)dx}$ gives the **area** between a positive curve $y = f(x)$, the **x-axis**, and the lines $x = a$ and $x = b$.
(We'll see why this works in a sec)



Find the area of the finite region between the curve with equation $y = 20 - x - x^2$ and the x-axis.



Factorise in order to find roots:

Therefore area between curve and x-axis is:

1.37 Integration: Definite Integrals and Areas

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Just for your interest…

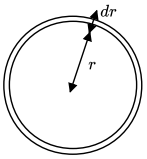
27

Why does integrating a function give you the area under the graph?

Part 1:

You're already familiar with the idea that gradient is the rate at which a quantity changes, and we consider an infinitesimally small change in the variable.
You could consider the gradient as the little bit you're adding on each time.

Here's some practical examples using formulae you covered in your younger years!

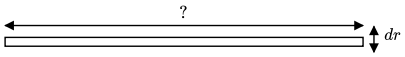


If you wanted to consider the rate at which the area of a circle increases with respect to the radius, consider a small change dr in the radius.
The change in area is an infinitely thin ring, which looks like you've drawn a circumference.
So what is this rate?
$A = \pi r^2$
$\therefore \frac{dA}{dr} = 2\pi r$



So to draw a shaded circle (which has area!), it's a bit like repeatedly drawing the circumference of a circle with gradually increasing radius.
Since the circumference is what the area is increasing by each time, the circumference is the gradient of the area of the circle, and conversely (by the definition of integration being the opposite of differentiation), the area is the integral of the circumference.

you can't add a length to an area
But by considering the infinitely thin width dr of the circumference you're drawing, it does have area!
$\frac{dA}{dr} = 2\pi r \quad \rightarrow \quad dA = 2\pi r dr$
i.e. the change in area, dA, is $2\pi r \times dr$
If we 'roll out' the added area we're adding each time, this forms a rectangle, whose length hopefully corresponds to the circumference of the circle:

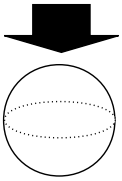


If the added area is $2\pi r dr$ and the thickness is dr, then the length is $\frac{2\pi r dr}{dr} = 2\pi r$ as expected.

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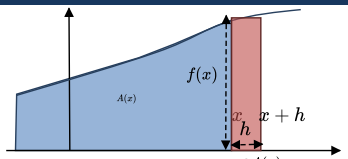
28

Part 2:



Consider the volume of a sphere:
$V = \frac{4}{3}\pi r^3$
$\frac{dV}{dr} = 4\pi r^2$

This works for a similar reason to before.



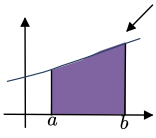
And the same principle applies to area under a graph. If $A(x)$ gives the area up to x, then what we're 'adding on each time' (i.e. the gradient) is sort of like drawing a line of length $f(x)$ at the right-most end each time (or more accurately, an infinitely thin rectangle of area $f(x) \times h$). Thus if $f(x)$ is the gradient of the area, then conversely the area is the integral of $f(x)$.

This gives us an **intuitive** sense of why it works, but we need a formal proof:

Using the diagram above, the area up to $x+h$, i.e. $A(x+h)$, is approximately the area up to x plus the added rectangular strip:
$$A(x+h) \approx A(x) + (f(x) \times h)$$
Rearranging:
$$f(x) \approx \frac{A(x+h)-A(x)}{h}$$
In the limit, as the rectangle becomes infinitely thin, this becomes an equality:
$$f(x) = \lim_{h\to 0} \frac{A(x+h)-A(x)}{h}$$
But we know from differentiating by first principles that:
$$A'(x) = \lim_{h\to 0} \frac{A(x+h)-A(x)}{h}$$
$$\therefore f(x) = A\prime(x)$$
And thus we have proven that the gradient of the area function is indeed $f(x)$ (and hence the integral of $f(x)$ the area).

One final Note: Why does $\int_b^a f(x)dx$ give the area between $x = a$ and $x = b$?

Since $f(x) = A\prime(x)$, the area function $A(x)$ is the integral of $f(x)$. Thus:
$$\int_b^a f(x)dx = [A(x)]_a^b = A(b) - A(a)$$



The area between a and b is the area up to b, i.e. $A(b)$, with the area up to a, i.e. $A(a)$, cut out. This gives $A(b) - A(a)$ as expected. Because we obtained $A(x)$ by an integration, we have a constant of integration $+c$. But because of the subtraction $A(b) - A(a)$, these constants in each of $A(b)$ and $A(a)$ cancel, which explains why the constant is not present in definite integration.

This is known as the Fundamental Theorem of Calculus.

1.37 Integration: Definite Integrals and Areas

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Test Your Understanding

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Edexcel C2 Jan 2013
Q9c

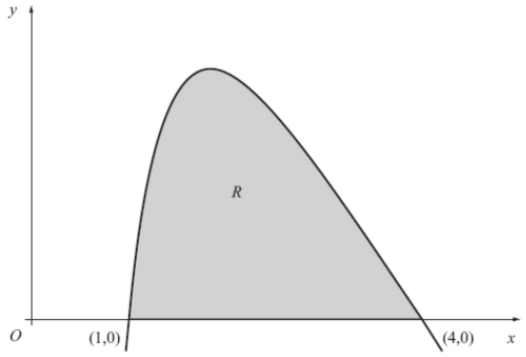


Figure 2

The finite region R, as shown in Figure 2, is bounded by the x-axis and the curve with equation
$y = 27 - 2x - 9\sqrt{x} - \frac{16}{x^2}, \qquad x > 0.$

The curve crosses the x-axis at the points $(1, 0)$ and $(4, 0)$.

(c) Use integration to find the exact value for the area of R. \hfill (6)

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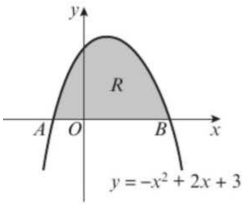
Exercise 13E – Y1 pages 299-300

30

8 The finite region R is bounded by the x-axis and the curve with equation $y = -x^2 + 2x + 3$, $x \geq 0$.
The curve meets the x-axis at points A and B.

a Find the coordinates of point A and point B.

b Find the area of the region R.



1.37 Integration: Definite Integrals and Areas

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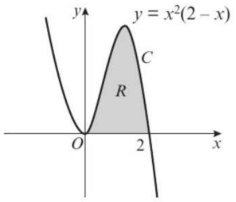
Exercise 13E – Y1 pages 299-300

31

9 The graph shows part of the curve C with equation $y = x^2(2 - x)$.

The region R, shown shaded, is bounded by C and the x-axis.

Use calculus to find the exact area of R. \hfill (5 marks)



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EXTENSION

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[MAT 2007 1H] Given a function $f(x)$, you are told that

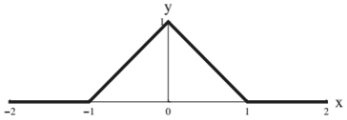
$$\int_0^1 3f(x)\,dx + \int_1^2 2f(x)\,dx = 7$$

$$\int_0^2 f(x)\,dx + \int_1^2 f(x)\,dx = 1$$

It follows that $\int_0^2 f(x)\,dx$ equals what?

[MAT 2011 1G]

A graph of the function $y = f(x)$ is sketched on the axes below:



What is the value of $\int_{-1}^1 f(x^2 - 1)\,dx$?

1.37 Integration: Definite Integrals and Areas

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‘Negative Areas’

33

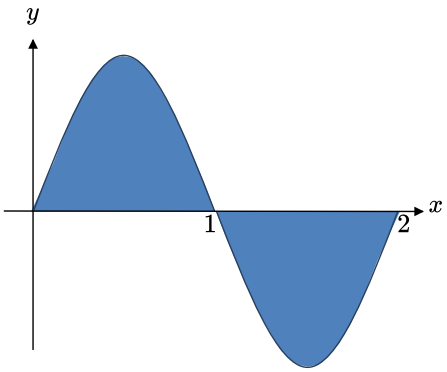
Sketch the curve $y = x(x-1)(x-2)$ (which expands to give $y = x^3 - 3x^2 + 2x$).
Now calculate $\int_0^2 x\,(x-1)\,(x-2)dx$. Why is this result surprising?

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Example

34

Find the total area bound between the curve $y = x(x-1)(x-2)$ and the x-axis.



Strategy:
Separately find the area between $x = 0$ and 1, and between 1 and 2. Treat any negative areas as positive.

1.37 Integration: Definite Integrals and Areas

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Test Your Understanding

35

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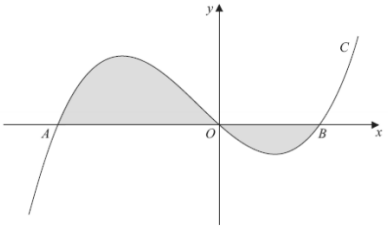


Figure 3 shows a sketch of part of the curve C with equation $y = x(x + 4)(x - 2)$.

The curve C crosses the x-axis at the origin O and at the points A and B.

(a) Write down the x-coordinates of the points A and B. \hfill (1)

The finite region, shown shaded in Figure 3, is bounded by the curve C and the x-axis.

(b) Use integration to find the total area of the finite region shown shaded in Figure 3. \hfill (7)

Figure 3

(a)

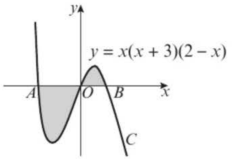
(b)

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Exercise 13F – Y1 pages 301-302

36

2 The graph shows a sketch of part of the curve C with equation $y = x(x + 3)(2 - x)$.



The curve C crosses the x-axis at the origin O and at points A and B.

a Write down the x-coordinates of A and B. \hfill (1 mark)

The finite region, shown shaded, is bounded by the curve C and the x-axis.

b Use integration to find the total area of the finite shaded region. \hfill (7 marks)

1.37 Integration: Definite Integrals and Areas

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Exercise 13F – Y1 pages 301-302

37

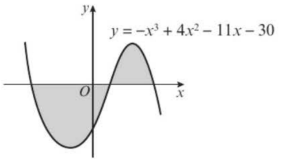
3 $f(x) = -x^3 + 4x^2 + 11x - 30$.

The graph shows a sketch of part of the curve with equation $y = -x^3 + 4x^2 + 11x - 30$.

a Use the factor theorem to show that $(x + 3)$ is a factor of $f(x)$.

b Write $f(x)$ in the form $(x + 3)(Ax^2 + Bx + C)$.

c Hence, factorise $f(x)$ completely.



d Hence, determine the x-coordinates where the curve intersects the x-axis.

e Hence, determine the total shaded area shown on the sketch.

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EXTENSION

38

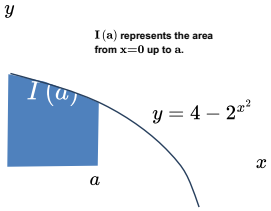
Extension

[MAT 2010 1I] For a positive number a, let

$$I(a) = \int_0^a \left(4 - 2^{x^2}\right) dx$$

Then $\frac{dI}{da} = 0$ when a is what value?

Hint: It's not actually even possible to integrate 2^{x^2}, but we can still sketch the graph. Reflect on what $\frac{dI}{da}$ actually means.



[STEP I 2014 Q3]

The numbers a and b, where $b > a \geq 0$, are such that

$$\int_a^b x^2 dx = \left(\int_a^b x dx\right)^2$$

(i) In the case $a = 0$ and $b > 0$, find the value of b.

(ii) In the case $a = 1$, show that b satisfies

$3b^3 - b^2 - 7b - 7 = 0$

Show further, with the help of a sketch, that there is only one (real) value of b that satisfies the equation and that it lies between 2 and 3.

(iii) Show that $3p^2 + q^2 = 3p^2q$, where $p = b + a$ and $q = b - a$, and express p^2 in terms of q. Deduce that $1 < b - a \leq \frac{4}{3}$

(iv)

Guidance for this problem on next slide.

1.37 Integration: Definite Integrals and Areas

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Guidance for Extension Problem 2

39

[STEP I 2014 Q3] The numbers a and b, where $b > a \geq 0$, are such that $\int_a^b x^2 dx = \left(\int_a^b x dx\right)^2$

(i) In the case $a = 0$ and $b > 0$, find the value of b.

(ii) In the case $a = 1$, show that b satisfies
$3b^3 - b^2 - 7b - 7 = 0$
Show further, with the help of a sketch, that there is only one (real) value of b that satisfies the equation and that it lies between 2 and 3.

(iii) Show that $3p^2 + q^2 = 3p^2q$, where $p = b + a$ and $q = b - a$, and express p^2 in terms of q.
Deduce that $1 < b - a \leq \frac{4}{3}$

This question was actually devised to address what happens when students misunderstand or mis-apply a "rule" of mathematics and *it turns out to give the right answer*. Part (i) starts you off gently: integrating both terms, squaring the RHS and solving very quickly gives $b = \frac{4}{3}$. Part (ii) develops in much the same way, but with a non-zero lower limit to the integrals, and we immediately see that the algebra gets much more involved. Importantly, it should be very clear that whatever expression materialises must have $(b - 1)$ as a factor (since setting $b = a$ would definitely give a zero area, thus trivially satisfying the given integral statement). This leads to the required cubic equation.

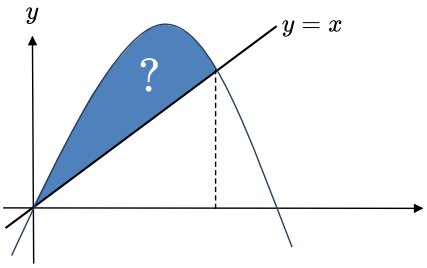
The final part of (ii) requires a mixture of different ideas (and can be done in a number of different ways). The most basic approach to demonstrating that a cubic curve has only one zero is to illustrate that both of its TPs lie on the same side of the x-axis (or to show there are no TPs). The popular *Change-of-Sign Rule* for continuous functions can be used to identify the position of this zero.

Having got you started with some simple lower limits, part (iii) develops matters more generally, and derives the (perhaps) surprising result that the exploration of this initial "stupid idea" requires b and a to be "not too far apart" to an extent that is easily identifiable.

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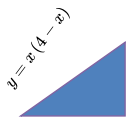
Areas between curves and lines

40



Determine the area between the lines with equations $y = x(4 - x)$ and $y = x$

How could we find the area between the line and the curve?



Start with the area under $y=x(4-x)$ up to the point of intersection, then subtract the area of the triangle to 'cut it out'.

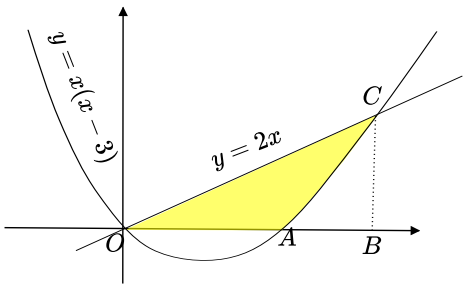
Click for animation >

1.37 Integration: Definite Integrals and Areas

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A Harder One

41



[Textbook] The diagram shows a sketch of the curve with equation $y = x(x-3)$ and the line with equation $y = 2x$.
Find the area of the shaded region OAC.

What areas should we subtract this time?

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Test Your Understanding

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Edexcel C2 May 2012
Q5

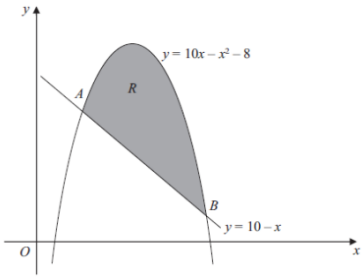


Figure 2 shows the line with equation $y = 10 - x$ and the curve with equation $y = 10x - x^2 - 8$.

The line and the curve intersect at the points A and B, and O is the origin.

(a) Calculate the coordinates of A and the coordinates of B. \hfill (5)

The shaded area R is bounded by the line and the curve, as shown in Figure 2.

(b) Calculate the exact area of R. \hfill (7)

a

b

Alternative Method:
If the top curve has equation $y = f(x)$ and the bottom curve $y = g(x)$, the area between them is:

$$\int_b^a (f(x) - g(x))dx$$

This means you can integrate a single expression to get the final area, without any adjustment required after.

1.37 Integration: Definite Integrals and Areas

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Exercise 13G – Y1 pages 304-306

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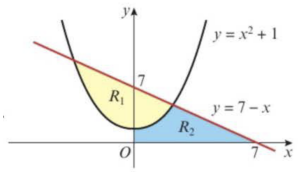
6 The diagram shows a sketch of part of the curve with equation $y = x^2 + 1$ and the line with equation $y = 7 - x$.

The finite region, R_1 is bounded by the line and the curve.

The finite region, R_2 is below the curve and the line and is bounded by the positive x- and y-axes as shown in the diagram.

a Find the area of R_1.

b Find the area of R_2.



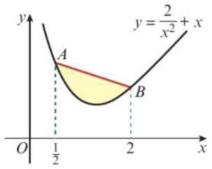
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Exercise 13G – Y1 pages 304-306

44

8 The diagram shows part of a sketch of the curve with equation $y = \frac{2}{x^2} + x$. The points A and B have x-coordinates $\frac{1}{2}$ and 2 respectively.

Find the area of the finite region between AB and the curve.



1.37 Integration: Definite Integrals and Areas

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Exercise 13G – Y1 pages 304-306

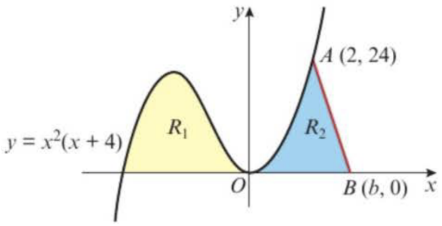
45

10 The sketch shows part of the curve with equation $y = x^2(x + 4)$. The finite region R_1 is bounded by the curve and the negative x-axis. The finite region R_2 is bounded by the curve, the positive x-axis and AB, where $A(2, 24)$ and $B(b, 0)$.

The area of R_1 = the area of R_2.

a Find the area of R_1.

b Find the value of b.



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EXTENSION

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- [MAT 2005 1A] What is the area of the region bounded by the curves $y = x^2$ and $y = x + 2$?

- [MAT 2016 1H] Consider two functions

  $f(x) = a - x^2$

  $g(x) = x^4 - a$

  For precisely which values of $a > 0$ is the area of the region bounded by the x-axis and the curve $y = f(x)$ bigger than the area of the region bounded by the x-axis and the curve $y = g(x)$?